ECE 310 Fall 2026
Lecture 5: Difference Equations and Block Diagrams

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Learning Objectives

- Interpret LTI systems defined by a Linear Constant-Coefficient Difference Equation (LCCDE).
- Identify whether a given LCCDE will yield a finite or infinite impulse response.



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Linear Constant Coefficient Difference Equations

- *Linear Constant Coefficient Difference Equations* (LCCDE) are a popular representation of LTI systems.
- The system output $y[n]$ is a linear combination of system inputs and outputs.

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j], \; 0 \leq K < \infty, \; 1 \leq M < \infty$$

y[n] = x[n] - x[n-1]

- K: number of output or *feedback* terms.
- M: number of input terms
- Couple of notes:
  1. Both K and M are finite while $M \geq 1$.
  2. We may express LCCDEs for non-causal systems as well.

Finite Impulse Response (FIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j], \; 0 \leq K < \infty, \; 1 \leq M < \infty$$

- If $K = 0$, i.e. no feedback terms, the system has a *finite-length impulse response* (FIR).
- In general, how can we find the impulse response of an FIR LTI system?

$$y[n] = \sum_{j=0}^{M-1} c_j x[n-j]$$

If $x[n] = \delta[n] \rightarrow y[n] = h[n] = \sum_{j=0}^{M-1} c_j \delta[n-j]$

Ex: $y[n] = x[n] - x[n-1]$

$\hookrightarrow h[n] = \delta[n] - \delta[n-1]$

Infinite Impulse Response (IIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- If $K > 0$, the system has an *infinite-length impulse response* (IIR).
- In general, how can we find the impulse response of an IIR LTI system?



Infinite Impulse Response (IIR) Systems

$$y[n] = \sum_{i=1}^{K} b_i y[n-i] + \sum_{j=0}^{M-1} c_j x[n-j],\ 0 \leq K < \infty,\ 1 \leq M < \infty$$

- If $K > 0$, the system has an *infinite-length impulse response* (IIR).
- In general, how can we find the impulse response of an IIR LTI system?
  - It's a bit more complicated.



Example 1: Determine the impulse response $h[n]$ of the below system.

$$y[n] = \frac{1}{3}y[n-1] + x[n]$$

Recall: $h[n]$ is the response $y[n]$ when $x[n] = \delta[n]$

$$\rightarrow y[n] = \frac{1}{3}y[n-1] + \delta[n]$$
$$h[n] = \frac{1}{3}h[n-1] + \delta[n]$$

By inspection: (by zero initial conditions)

$$h[0] = \frac{1}{3}h[-1] + \delta[0] = 1$$
$$h[1] = \frac{1}{3}h[0] + \delta[1] = \frac{1}{3}$$
$$h[2] = \frac{1}{3}h[1] + \delta[2] = \frac{1}{9}$$
$$h[3] = \frac{1}{3}h[2] = \frac{1}{27}$$

By inspection

$$h[n] = \left\{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \cdots\right\}$$
$$= \left(\frac{1}{3}\right)^n u[n]$$

Infinite Impulse Response (IIR) Systems

Example 2: Determine the impulse response $h[n]$ of the below system.

$$y[n] = \frac{1}{3}y[n-1] + x[n] - 3x[n-1] + x[n-2]$$



Infinite Impulse Response (IIR) Systems

Example 2: Determine the impulse response $h[n]$ of the below system.

$$y[n] = \frac{1}{3}y[n-1] + x[n] - 3x[n-1] + x[n-2]$$

Consider another system: $\hat{y}[n] = \frac{1}{3}\hat{y}[n-1] + \hat{v}[n]$

($\hat{y}[n]$: output; $\hat{v}[n]$: input)

→ If $\hat{v}[n] = \delta[n]$, $\hat{y}[n] = \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n]$ (from previous slide)

Example 2: Determine the impulse response $h[n]$ of the below system.

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Consider another system: $\hat{y}[n] = \frac{1}{3}\hat{y}[n-1] + \hat{v}[n]$

↑ output ↗ ↑ input

→ If $\hat{v}[n] = \delta[n]$, $\hat{y}[n] = \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n]$ (from previous slide)

→ If $\hat{v}[n] = \delta[n] - 3\delta[n-1] + \delta[n-2]$, $\hat{y}[n] = \hat{v}[n] * \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n] - 3\left(\frac{1}{3}\right)^{n-1} u[n-1] + \left(\frac{1}{3}\right)^{n-2} u[n-2]$

$\left(\hat{y}[n] = \frac{1}{3}\hat{y}[n-1] + \delta[n] - 3\delta[n-1] + \delta[n-2]\right)$

$-3\hat{h}[n-1]$ $\hat{h}[n-2]$

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output input

→ If $\hat{v}[n] = \delta[n]$, $\hat{y}[n] = \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n]$ (from previous slide)

→ If $\hat{v}[n] = \delta[n] - 3\delta[n-1] + \delta[n-2]$, $\hat{y}[n] = \hat{v}[n] * \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n] - 3\left(\frac{1}{3}\right)^{n-1} u[n-1] + \left(\frac{1}{3}\right)^{n-2} u[n-2]$

$\left(\hat{y}[n] = \frac{1}{3}\hat{y}[n-1] + \delta[n] - 3\delta[n-1] + \delta[n-2]\right)$ ← same as this!

$-3\hat{h}[n-1]$ $\hat{h}[n-2]$

→ If $x[n] = \delta[n]$, $y[n] = \frac{1}{3}y[n-1] + \delta[n] - 3\delta[n-1] + \delta[n-2]$,

↳ $y[n] = h[n] = \left(\frac{1}{3}\right)^n u[n] - 3\left(\frac{1}{3}\right)^{n-1} u[n-1] + \left(\frac{1}{3}\right)^{n-2} u[n-2]$

Infinite Impulse Response (IIR) Systems

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($\hat{y}[n]$: output; $\hat{y}[n-1]$: output; $\hat{v}[n]$: input)

→ If $\hat{v}[n] = \delta[n]$, $\hat{y}[n] = \hat{h}[n] = \left(\frac{1}{3}\right)^n u[n]$ (from previous slide)

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$\left(\hat{h}[n] = \frac{1}{3}\hat{y}[n-1] + \delta[n] - 3\delta[n-1] + \delta[n-2]\right)$

